

Special Article

RO-ILS: Radiation Oncology Incident Learning System: A report from the first year of experience



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Abstract

Purpose: Incident learning is a critical tool to improve patient safety. The Patient Safety and Quality Improvement Act of 2005 established essential legal protections to allow for the collection and analysis of medical incidents nationwide.

Methods and materials: Working with a federally listed patient safety organization (PSO), the American Society for Radiation Oncology and the American Association of Physicists in Medicine established RO-ILS: Radiation Oncology Incident Learning System (RO-ILS). This paper provides an overview of the RO-ILS background, development, structure, and workflow, as well as examples of preliminary data and lessons learned. RO-ILS is actively collecting, analyzing, and reporting patient safety events.

Results: As of February 24, 2015, 46 institutions have signed contracts with Clarity PSO, with 33 contracts pending. Of these, 27 sites have entered 739 patient safety events into local database space, with 358 events (48%) pushed to the national database.

Conclusions: To establish an optimal safety culture, radiation oncology departments should establish formal systems for incident learning that include participation in a nationwide incident learning program such as RO-ILS.

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Introduction

The Institute of Medicine (IOM) report “To Err Is Human: Building a Safer Health System” defined the reduction of harm to patients as a critical need. One of the central strategies for improvement was the identification of and learning from errors through large-scale reporting systems.¹ There are data to support the efficacy of such an approach. Annual national hospital surveys from the Agency for Healthcare Research and Quality (AHRQ) have demonstrated that higher levels of near-miss incident reporting are significantly correlated with fewer patient safety-related adverse events.²

Incident learning is the process of reporting and analyzing safety events and designing and implementing interventions intended to prevent future safety events from occurring.³ This strategy of reporting, analysis, and improvement has played a key role in the reduction of fatal airline accidents⁴ and is in widespread use in the nuclear power industry.⁵

The AHRQ established common definitions and reporting formats that include 3 classifications of patient safety events. Incidents are defined as patient safety events that reached the patient, whether or not there was harm involved. Near misses (or close calls) are patient safety events that did not reach the patient. Unsafe conditions are circumstances that increase the probability of a patient safety event occurring.⁶

Within radiation oncology, there is evidence that incident learning results in fewer serious safety events over time.⁷⁻⁹ For example, 1 radiation oncology group reports a >30% reduction in safety events attributed to interventions prompted by incident analysis.⁹ It has also been shown to improve the culture of safety within a department as measured with validated indicators.¹⁰

Methods and materials

In 2011, the American Society for Radiation Oncology (ASTRO) partnered with the American Association of Physicists in Medicine (AAPM) to establish a nationwide incident learning system as an opportunity to share learning across all radiation oncology institutions. The system is called RO-ILS: Radiation Oncology Incident Learning System (RO-ILS). RO-ILS forms a key component of Target Safely, the plan developed by ASTRO to improve quality and safety in radiation oncology. The mission of RO-ILS is to facilitate safer and higher-quality care in radiation oncology by providing a mechanism for shared learning in a secure and nonpunitive environment. This reports shares the experience from the first year of operation.

Response to error and just culture

As a result of the IOM report, the US Congress and other federal stakeholders began to strategize ways to improve safety. One result was the enactment of a federal statute, the Patient Safety and Quality Improvement Act of 2005 (PSQIA).

Health care providers have many concerns surrounding the willing admission of mistakes, including a historically punitive paradigm toward those involved in errors, coupled with concerns about legal actions.^{11,12} For example, in an attempt to address errors, some managers and providers take the position that the most appropriate response is to punish the health care providers involved, believing that this will prevent recurrence. Some also believe punishment may deter other providers. Often, those who suffer from medical errors believe that justice involves remuneration or punishment through the court system, with the belief that such activities may provide healing.

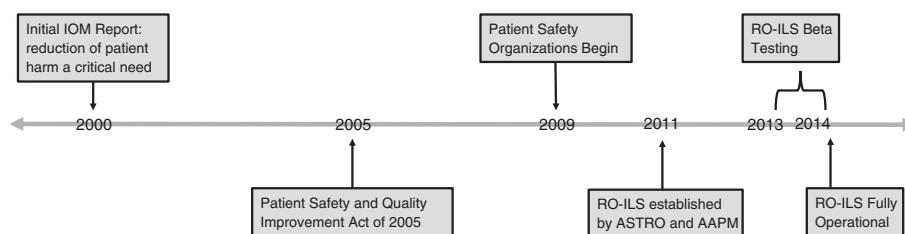


Figure 1 Timeline for development of RO-ILS program. AAPM, American Association of Physicists in Medicine; ASTRO, American Society for Radiation Oncology; IOM, Institute of Medicine; RO-ILS, Radiation Oncology Incident Learning System.

However, a punitive paradigm and litigation often do not prevent mistakes and cannot offer real solutions. This is supported by a well-established body of work.¹³ Furthermore, potentially unsafe processes (which account for the majority of errors in complex systems) remain unidentified. The long-term effects of punitive responses on medical errors may in fact degrade patient safety.

Patient safety organizations

The PSQIA was enacted in light of the above insight and history. PSQIA allows for the creation of patient safety organizations (PSOs). The AHRQ issued final regulations in 2009 (42 CFR Part 3) establishing guidance for PSOs and providers. There are currently 83 listed PSOs, ranging from individual hospital-based PSOs to state-wide PSOs to medical specialty-sponsored PSOs such as “Wake up Safe” for pediatric anesthesiology. The PSO program is voluntary and allows health care providers to work with federally certified PSOs to improve quality and safety. In turn, the data and analysis shared and conducted between providers and PSOs are subject to strong federal protections. This patient safety work product (PSWP) is privileged and confidential under statute, which allows for the necessary deep analysis of errors in a safe and protected environment.

In June 2011, ASTRO approved a proposal to establish a national radiation oncology-specific incident learning system. Figure 1 shows a timeline of this program in the context of other developments on a larger scale. ASTRO partnered with the AAPM to develop RO-ILS, which is the only medical specialty society-sponsored incident learning system for radiation oncology. ASTRO serves as the gateway for RO-ILS participation and contracted with Clarity PSO, a federally certified and listed PSO, to develop and administer RO-ILS. Federal regulations require that there be a direct relationship between the health care provider and the PSO. Although ASTRO and AAPM sponsor the program and have a relationship with Clarity PSO, there is also a formal and separate contractual agreement between Clarity PSO and radiation oncology providers who choose to participate.

Participation in the PSO program requires active 2-way reporting between the provider and the PSO. Clarity PSO

developed the RO-ILS portal using the architecture of its Healthcare *SafetyZone* Portal, an Internet-based system used by a range of health specialties. Radiation oncology providers who contract with Clarity PSO gain access to this secure portal to submit information on incidents, near misses, and unsafe conditions. Data submitted via RO-ILS and the *SafetyZone* Portal are protected under the PSQIA statute. Additional information, including a user’s guide and frequently asked questions, is available on the ASTRO website.¹⁴

RO-ILS development and data structure

The structure of RO-ILS is based largely on an AAPM 2012 report that describes recommendations and data structures for incident learning systems in radiation oncology.¹⁵ This report was initiated at the ASTRO/AAPM “Call to Action” safety meeting held in Miami, Florida, in 2010.¹⁶ In April 2012, ASTRO and AAPM began the development of what would become RO-ILS. A year later, a contract was signed with Clarity PSO, and the development of the web portal began in earnest.

The web portal contains a number of data elements, the most important of which is the event narrative. Events are primarily classified as incidents that reached the patient, near misses, or unsafe conditions. Other data elements include information about the patient, treatment goals and techniques (all protected under the PSO structure), workflow descriptors, event severity, hardware and software involved, and causal factors. The web portal also asks specifically, “How was the event discovered?” and “What changes, if any, has the facility made in response to the report?” Most RO-ILS data elements are optional. Certain elements must be reported given AHRQ’s federal requirements. Often, the most useful information entered into an event report is the narrative description. The narrative describes the details of the event and is often rich in information and context. Users are encouraged to input as much information as possible in each report.

All data entered reside on servers maintained by Clarity PSO. This eliminates the need for local information technology support. The data stay within the sequestered database for that facility, which enables reports to include identifying information. Most institutions elect to include

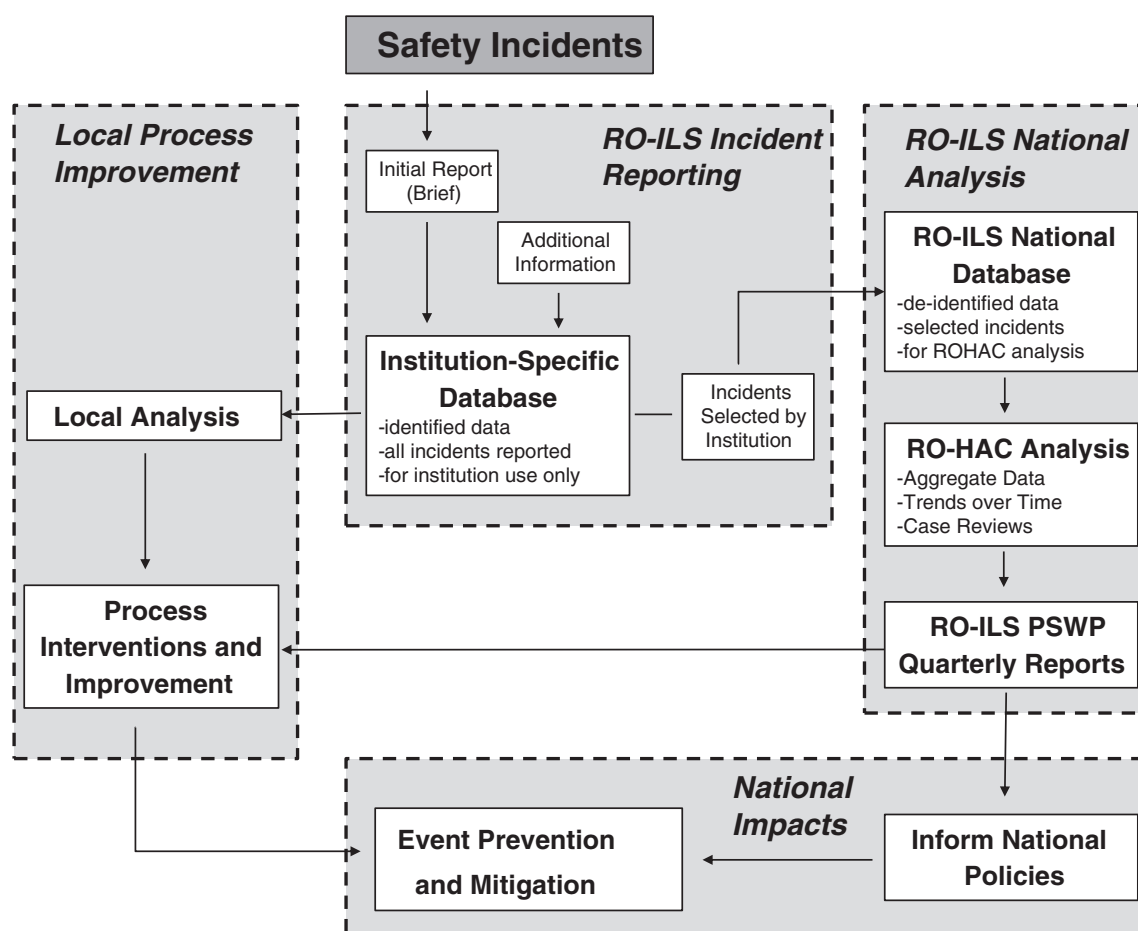


Figure 2 Current workflow for Radiation Oncology Incident Learning System (RO-ILS). PSWP, patient safety work product; RO-HAC, Radiation Oncology Health Advisory Council.

identifying information to facilitate follow-up by local staff, to learn from patient safety events, and to provide feedback. The web portal has modules to assist institutions with searching, reviewing, and analyzing their events. The system is compliant with all Health Insurance Portability and Accountability Act (HIPAA) requirements.

RO-ILS beta testing

The goals of RO-ILS beta testing were to assess the precontracting/contracting experience, the website interface and data elements, and the reports produced by the Radiation Oncology Health Advisory Council (RO-HAC), a group of radiation oncology professionals who assist Clarity PSO to analyze and report on the data that are submitted via RO-ILS. A 3-phase evaluation approach was used across beta sites between September 2013 and November 2014. There were 14 beta sites, including 2 freestanding practices, 2 community-based hospital practices, and 10 academic practices.

The first phase of evaluation focused on the process by which a contract is established between the clinical facility

and Clarity PSO. Structured telephone interviews of staff at beta test sites were conducted. Emerging themes included challenges shepherding the contract through respective institutions, concerns about data security, and insufficient information on the contracting process in educational materials. As a result of this phase, Clarity PSO made modifications to the contract by addressing information technology and data security issues. Additionally, educational resources were updated to include more direction about the contracting process. These interventions compressed the contracting process from 8 to 12 weeks to less than 8 weeks.

The second phase, conducted from May to July 2014, targeted the users' experience with the data elements. A survey that addressed each data element was sent to the beta testing sites. As a result of feedback, data elements were reorganized on the basis of priority for analysis. Additionally, equipment and software data elements were restructured, and the contributing factors section was optimized.

The third phase, conducted in November 2014, focused on the quality and usefulness of the quarterly aggregate reports. Initial reports during beta testing included an

aggregate report card, which provided a high-level summary of data elements; an analysis and commentary section, which provided a deeper analysis of events, including lessons learned; and a section on improving the use of RO-ILS, which provided tips on how to maximize the user experience. Only beta testers received the initial aggregate quarterly report. Given the limited data, customized institutional reports were not created. Beta testers were surveyed regarding each report section, including an assessment of the impact to the department. The results focused on report formatting recommendations and the addition of more detail to root cause analysis. These recommendations were instituted in subsequent reports.

In summary, the evaluation of the beta testers provided meaningful data that helped improve RO-ILS. Means for continued progress will include an annual survey (from contracting through the assessment of reports) to further optimize RO-ILS.

RO-ILS current operation

RO-ILS workflow can be usefully separated into events within the reporting department and processes that happen after the data are uploaded to the national PSO for review. Figure 2 summarizes RO-ILS current workflow.

Contract with PSO

The reporting facility first enters into a contract with Clarity PSO. This establishes mandated legal protections that enable the reporting facility to have a protected database within the PSO information system. As currently constituted, there is no cost for participation in RO-ILS.

Workflow within a facility

Initial event entry captures key data elements, including a narrative, and is designed to require roughly 1 minute to enter. Once the initial event is entered, designated staff are automatically emailed by the system. It is expected that supervisory staff who receive event notification will investigate further and add supplementary information. The event report may be accessed and updated multiple times as understanding evolves. Data elements requested for each event are described elsewhere in this paper. Best practices for event entry include the following:

- Narratives should be sufficiently detailed such that the event is understandable to both internal and external potential reviewers. Note that in addition to the initial narrative entered by a frontline provider, the review section of the web portal allows for the entry of further description as the event is investigated.
- The facility's corrective action should be described to allow others to learn not only what happened but how such events may be prevented or mitigated.

- Categorization choices should be selected completely and carefully so that data can be effectively compiled and analyzed for trends. The institution itself can perform such analysis to understand the patterns in their data. Trends can also be analyzed in aggregate by Clarity PSO.

When local reviewers within the institution have finished updating the report, they have the option of marking the data for upload to the national PSO database. The decision about which events to push to the national database is left to the reporting institution and varies by institution. Participating institutions are encouraged to submit any patient safety event that may improve quality and safety, including near misses.

Aggregate review process and quarterly reports

A group of radiation oncology professionals, selected by ASTRO and AAPM, work under contract with Clarity PSO to review the anonymized event reports and create summary documents. This group, RO-HAC, is composed of 8 people who serve staggered terms and represent physicians, physicists, dosimetrists, and administrators.

The RO-HAC members review the reports, rank them for importance and interest, look for commonalities and trends, and then summarize the results with statistics of key categories. This analysis, along with the individual department reports, is designated as PSWP and represents the primary output of RO-ILS. Importantly, this output provides succinct descriptions of the most salient reports and lessons learned.

On the basis of RO-HAC analysis, a quarterly report is generated and sent to the participating institutions. Quarterly reports are also posted to the community at large on the website (astro.org/roils under the "Reports" tab). The quarterly reports include descriptive statistics, for example, the number of reports by category (incident vs near miss vs unsafe event), as well as the step in the workflow where an event originated or was detected. The reports also include an analysis and commentary section, as well as case studies drawn from the reports of that quarter. These reports are designated as protected PSWP, an important distinction as the program evolves and institutions begin to receive customized reports that include institutional benchmarking data.

Results

First year of RO-ILS: Summary of data and lessons learned

As of February 24, 2015, 46 institutions had signed contracts with Clarity PSO, with 33 contracts pending. Of these, 27 sites had entered 739 patient safety events into local database space. The median number of events per site was 14 (range, 1-182). Of the 739 total events, 358 (48%)

were pushed to the national database for RO-HAC analysis. Between February 20 and December 31, 2014, a total of 220 events were analyzed by the RO-HAC, including events marked as patient incidents (44%), near misses (33%), and unsafe conditions (23%).

On the basis of a cluster of cases during the first quarter, RO-HAC chose to emphasize isocenter identification as a topic for discussion. Wrong isocenter identified was a potential systematic error in 15% of events. At least 4 distinct originating factors were identified. In several cases, the error passed through early review steps before being caught, which emphasizes the principles of layered security.

Across the data collected to date, the treatment planning and pretreatment review/verification workflow steps were consistently the most commonly identified points at which an error occurred (33% of events), with a caveat regarding missing data discussed below. This is in line with other published data showing treatment preparation to be the most common point of error origination.⁹ In reviewing submitted cases, communication, changes to plans, and training/education emerged as common themes that may drive errors in treatment planning/preparation.

To date, 27 patient safety events have been submitted to RO-ILS as having a dosimetric effect, including 1 >100% dose deviation, 2 deviations of 25% to 100%, 8 deviations of 5% to 25%, and 16 deviations of <5%. These 27 reports represent 12% of cases with some dose deviation. However, on further review by the RO-HAC, it was determined that 7 of these events were potential dose deviations, scored as if the events had reached the patient when in fact the event had not reached the patient. The remaining 20 were confirmed dose deviations. This analysis led to significant discussion regarding defining actual versus potential error and improving the definition of events that “reach the patient.” This work is ongoing as RO-HAC seeks to produce an easily generalizable radiation oncology-specific definition of events that reach the patient.

The RO-ILS data to date show nearly 3 times as many patient safety events involving 3-dimensional conformal radiation therapy as intensity modulated radiation therapy (IMRT; 29% vs 11%). To put this in context, the proportion of external beam radiation therapy delivered as IMRT, as measured by Medicare coding, has increased rapidly from 0% in 2002 to 54% in 2010.¹⁷ The patterns in the RO-ILS data are congruent with published research that suggests lower error rates in IMRT than with 3-dimensional/conventional techniques.¹⁸ With additional data, future work should help identify patterns and suggest interventions to mitigate techniques or modalities that are shown to have higher risks. Future work should not only focus on the frequency of error but also include an analysis of possible differential harm.

The case studies in quarterly reports serve to illustrate basic safety principals and process improvements. Such cases may illustrate novel process improvements or, more commonly, re-emphasize established safety practices. Given that there is often a considerable gap between what is considered best practice and what is done routinely in the clinic,¹⁹ process improvements are important.

Discussion

Plans for future development

For RO-ILS to reach its full potential for promoting meaningful changes in patient safety, participation must increase. The IOM report specifically discusses the power of nationwide systems, and the PSQIA was established to enable and protect nationwide reporting. In addition to immaturity, a major shortcoming of RO-ILS to date is the high frequency of items unanswered or marked “unsure.” Analysis is hobbled for items with high nonresponse and unsure rates. For example, “unanswered/not sure” was the most commonly identified response for event characterization (59%) and for workflow step at which the event occurred (60%). Although initial event entry requires only a minute or so, follow-up including supplemental information takes substantially longer. There is a fine balance between optimizing input for ease/speed while maintaining a broad and detailed capture of data elements. RO-ILS currently is completely dependent on manual data entry for each item. Advances in technology, including working with vendors to integrate incident reporting into standard clinic software, have the potential to improve the process.

As indicated in the discussion of cases with dose deviations above, another hurdle is a potential disconnect between the classification of events by the reporting team and by the RO-HAC analysis. As RO-ILS grows and the volume of patient safety events to be reviewed expands, RO-HAC may be unable to individually review all cases and will be dependent on correct classification for prioritization and robust aggregate analysis. In an attempt to reduce the proportion of items marked unsure or unanswered, and to improve the classification of events, there is an ongoing process to improve the RO-ILS data elements.

User training is another important strategy to improve event entry. As institutions begin a RO-ILS relationship, Internet conference-based training is supplied by Clarity PSO. In addition, the PSWP quarterly reports include suggestions for improving the use of RO-ILS. Training will be evaluated and modified as needed to improve event entry. Clarity PSO also periodically requests user feedback and hosts ongoing discussions with RO-ILS users to improve the process.

Other organizations have also recognized the importance of training for incident learning. The AAPM, with support by ASTRO, organized a February 2015 hands-on workshop titled, "Incident Learning Systems and Root Cause Analysis for Safer Radiation Oncology." There were traditional lectures and an interactive aspect to the workshop. More than 80 participants learned through simulated audiovisual incident cases, including hands-on event entry into the RO-ILS interface. A total of 93 reports were entered based on the simulated events and are being analyzed for consistency in classification choices. The lectures will be available through the AAPM Virtual Library.

Conclusions and recommendations

Incident learning through RO-ILS represents a powerful opportunity to potentially address patient safety and improve outcomes. PSQIA established the required legal protections to collect, analyze, and learn from patient safety events nationwide. ASTRO and the AAPM, working with Clarity PSO, have established RO-ILS to enhance patient safety by facilitating nationwide incident learning and building awareness of practice risks in radiation oncology. RO-ILS, although in its infancy, is actively collecting, analyzing, and reporting incident learning data. We recommend that all radiation oncology departments establish formal systems for incident learning that include participation in a nationwide incident learning program.

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